NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

28 July 2026

233 verified · 416 raw · 5 sources

233	2	231	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	211	Media
TechCrunch AI	10	Media
HackerNews	9	Community
MIT Tech Review	3	Media
The Verge AI	2	Media

VERIFIED INTELLIGENCE — 2 stories

VERI

Building the enterprise environment for agentic AI

MIT Tech Review +1 source 80%

For the enterprise, the promise of agentic AI is much more than just a better chatbot. It is software agents that execute business tasks end-to-end across people, business workflows, data, and systems. The platform best-suited to run agents is built with proper CPU capacity,...

https://www.technologyreview.com/2026/07/27/1140668/building-the-enterprise-environment...

VERI

Hierarchical Grading in Large Language Models

ArXiv CS.AI +1 source 80%

arXiv:2607.22757v1 Announce Type: cross Abstract: We introduce Graded Large Language Models (GLLMs), an algebraic framework that equips the representation space of a transformer with a grading and propagates the induced weighted scalar action through embeddings, self-attention,...

https://arxiv.org/abs/2607.22757

CONFIRMED SIGNALS — 231 stories

CONF

Show HN: FeyNoBg – Automatic background removal model and training library

HackerNews 60% HN 101

Hey HN, I’m Shreyash from Feyn. We help companies build custom models from their data. Today, we’re releasing FeyNoBg, an automatic background removal model. Alongside it, we're open-sourcing NoBg, the Python library we built to train and run it. Try the model here:...

https://usefeyn.com/blog/feynobg/

CONF

Show HN: Watch 14-Byte AI "brains" attempt to solve a 2D maze (Its hard)

HackerNews 50% HN 23

Hey HackerNews, I built this project over the last few weeks as a palette cleanser from a failed game launch. I wanted to learn a bit about AI/Neural-Networks and naively thought I could build a tiny maze-solving AI in a weekend with a 100% solve rate. Well - I couldn't, but I...

https://con-dog.github.io/MINIMIO-PUBLIC-FRONTEND/

CONF

Show HN: Running PrismML's Bonsai inside DRAM by breaking DDR4 timing rules

HackerNews 50% HN 22

The excitement surrounding PrismML’s 1-bit/ternary Bonsai models has the industry closely watching how smartphone giants, particularly Apple, will implement LLMs on edge devices. Moving AI on-device is a brilliant and necessary strategy. It ensures absolute user privacy in...

https://news.ycombinator.com/item?id=49019271

CONF

Show HN: Call Me, your AI agents ring you with no extra carrier costs

HackerNews 50% HN 5

Using chat interfaces often spawns many different thoughts, and while the AI and it's agents are okay at keeping track of it, I'm not always there. I needed to be able to see where my thoughts diverged, and retrace them back to the source. While mulling over a better interface,...

https://github.com/radres/call-me

CONF

Show HN: SeaTicket – AI agent that resolve GitHub and Discord issues

HackerNews 50% HN 4

https://seaticket.ai/ After maintaining Seafile, open-source file-sync software, since 2012. Somewhere across those fourteen years, "go check if someone already reported this" turned into one of the most common lines in our team chat. Because the same bug tended to show up...

https://news.ycombinator.com/item?id=49078625

CONF

Show HN: Building a new game everyday with AI. Day #106 Zombie Survival Training

HackerNews 50% HN 3

I'm using AI (mostly Claude) to create/publish a new video game every day This is day 106, and my first stab at the zombie/survival genre. Most of the games I build with just a few prompts, but this one I used almost an entire 5-hour window on the $20/month pro subscription (in...

https://gamevibe.us/106-zombie-survival-training

CONF

Show HN: PromptTrace – Free hands-on labs to practice hacking LLMs

HackerNews 50% HN 3

No summary — see link for full article.

https://prompttrace.airedlab.com

0 CONF

Show HN: _done – Expanding AI Capabilities

HackerNews 50% HN 2

Actions for AI. Real-world tasks your agent can't do alone. Pay per use with x402 - no accounts. no logins. no keys.

https://underscoredone.com/

1 CONF

Show HN: FileForge Finder: Local file search with AI-optimized export

HackerNews 50% HN 2

Hope this sparks some conversation. Interested to get the community's take on an app we use internally to find files and provide targeted file content to chatbots. Biggest gains in search are for windows users but I am interested to get feedback from the spotlight users. With...

https://news.ycombinator.com/item?id=49076401

2 CONF

Anthropic’s Dario Amodei responds: doesn’t oppose open-weight models, but fears Chinese AI

TechCrunch AI 50%

Anthropic founder and CEO Dario Amodei made his views clear about open-weight models and China's growing AI capabilities.

https://techcrunch.com/2026/07/27/anthropics-dario-amodei-responds-doesnt-oppose-open-w...

3 CONF

Satya Nadella says companies that trust one AI for everything may not survive

TechCrunch AI 50%

Companies without their own models — or without a layer of AI infrastructure known as AI gateways to separate their prompts from the model itself — will be in trouble, Nadella says.

https://techcrunch.com/2026/07/27/satya-nadella-says-companies-that-trust-one-ai-for-ev...

4 CONF

PSA: Your Claude shared chats and Artifacts may have ended up on Google

TechCrunch AI 50%

The issue appears to have originated from Claude’s “share chat” feature, which allows users to create links that enable anyone with the assigned URL view a conversation or project.

https://techcrunch.com/2026/07/27/psa-your-claude-shared-chats-and-artifacts-may-have-e...

5 **CONF**

OpenAI’s Hugging Face breach has reignited the debate over alignment and control

TechCrunch AI **50%**

OpenAI's Hugging Face breach has reignited debate over AI alignment and control, exposing competing views on whether increasingly capable AI should be better aligned, better contained, or both.

https://techcrunch.com/2026/07/27/openais-hugging-face-breach-has-reignited-the-debate-...

6 **CONF**

Threads users can now chat with Meta AI in their DMs

TechCrunch AI **50%**

Meta on Monday said it is rolling out its Meta AI chatbot within Threads' DMs, giving users a way to chat with the AI assistant.

https://techcrunch.com/2026/07/27/threads-users-can-now-chat-with-meta-ai-in-their-dms/

7 **CONF**

Google’s AI search is rapidly becoming the default, new data shows

TechCrunch AI **50%**

Google’s AI Overviews now appear in 43% of searches, underscoring how quickly AI-generated answers are becoming the default way people discover information online.

https://techcrunch.com/2026/07/27/googles-ai-search-is-rapidly-becoming-the-default-new...

8 **CONF**

Power up your AI infrastructure! A first look at the Smart Systems Stage agenda at TechCrunch Disrupt 2026

TechCrunch AI **50%**

At TechCrunch Disrupt 2026, the Smart Systems Stage will be where energy, infrastructure, and technology collide, covering everything from fusion breakthroughs to the grid strain AI is putting on the entire economy.

https://techcrunch.com/2026/07/27/power-up-your-ai-infrastructure-a-first-look-at-the-s...

9 **CONF**

Ilya Sutskever’s Safe Superintelligence partners with Nvidia to scale its AI research

TechCrunch AI **50%**

After two years in stealth, Safe Superintelligence has announced a long-term partnership with Nvidia as it prepares to scale to its next phase.

https://techcrunch.com/2026/07/27/ilya-sutskevers-safe-superintelligence-partners-with-...

0 **CONF**

Enigma raises $71M to make controlling a robot as easy as adjusting the volume

TechCrunch AI **50%**

The massive seed round was led by Index Ventures and Ribbit Capital, with participation from Sarah Guo's Conviction Partners.

https://techcrunch.com/2026/07/27/enigma-raises-70m-to-make-controlling-a-robot-as-easy...

1 **CONF**

One fallen power line exposed a growing AI data center problem — here’s how to fix it

TechCrunch AI **50%**

A close call in Northern Virginia revealed just how poorly data centers respond to grid disruptions. Here's how to fix the problem.

https://techcrunch.com/2026/07/25/one-fallen-power-line-exposed-a-growing-ai-data-cente...

2 **CONF**

OpenAI called the Hugging Face attack unprecedented. But we’ve been here before.

MIT Tech Review **50%**

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. Reading OpenAI’s account last week of how some of its models broke their containment and hacked into the computer systems of Hugging Face,...

https://www.technologyreview.com/2026/07/27/1140836/openai-hugging-face-attack-precedent/

3 CONF

Closing the data loop in AI-driven drug discovery

MIT Tech Review **50%**

Drug discovery is a high-cost, high-risk endeavor that is under growing pressure from a market increasingly defined by first-mover advantage. Since the 1950s, the cost of developing new pharmaceuticals has roughly doubled every nine years—a phenomenon known as Eroom's Law....

https://www.technologyreview.com/2026/07/27/1139667/closing-the-data-loop-in-ai-driven-...

4 CONF

Nvidia, Microsoft launch open AI security alliance — without OpenAI, Google, or Anthropic

The Verge AI **50%**

Nvidia on Monday said it is joining forces with Microsoft, SpaceX, IBM, and other tech companies to build and share open-source AI security tools. The new Open Secure AI Alliance said open tools are required to effectively defend against attacks from frontier models. The...

https://www.theverge.com/ai-artificial-intelligence/971281/nvidia-open-secure-ai-allian...

5 CONF

TRE: Training-Free Hallucination Detection for Diffusion Language Models

ArXiv CS.AI **50%**

arXiv:2607.22661v1 Announce Type: new Abstract: Diffusion large language models (D-LLMs) have recently gained increasing attention, yet their reliability is significantly hindered by the hallucination problem. Existing hallucination detection approaches for D-LLMs mainly follow...

https://arxiv.org/abs/2607.22661

6 CONF

Stress-Testing EEG Foundation Models for Clinical Decoding: Dataset Identity and Targeted Negative Controls

ArXiv CS.AI **50%**

arXiv:2607.24519v1 Announce Type: cross Abstract: Pretrained EEG foundation models are increasingly proposed for clinical decoding, but their transfer across populations and robustness to negative controls remain unclear. We benchmark six models (LaBraM, EEGMamba, CBraMod, REVE,...

https://arxiv.org/abs/2607.24519

7 CONF

Same Question, Different Answers: Evaluating LLM Reliability Beyond Accuracy

ArXiv CS.AI **50%**

arXiv:2607.22554v1 Announce Type: new Abstract: Large language models (LLMs) often achieve strong accuracy on benchmarks, yet it remains unclear how reliably they apply this knowledge when the same question is phrased in different but equivalent ways. In this work, we study how...

https://arxiv.org/abs/2607.22554

8 CONF

DeepLens Diagnosis Agent: Agentic Workflow Design Lets a Small Reasoning Model Compete with Frontier LLMs

ArXiv CS.AI **50%**

arXiv:2607.22555v1 Announce Type: new Abstract: Medical diagnosis is a multi-stage process: extract facts, consult knowledge, generate a differential analysis, and select the best diagnosis with explanations. Frontier LLMs are strong generalists, but single-shot prompting often...

https://arxiv.org/abs/2607.22555

9 **CONF**

Masked Distillation: Internalizing the Chain-of-Thought in Language Models

ArXiv CS.AI **50%**

arXiv:2607.22629v1 Announce Type: new Abstract: Large Reasoning Models (LRMs) produce long, explicit chains of intermediate steps before generating a final answer at inference time. These intermediate traces dominate latency, memory usage, and serving cost, even though the final...

https://arxiv.org/abs/2607.22629

0 **CONF**

MemChain: Learning Interpretable Memory Traces for Memory-Augmented LLM Agents

ArXiv CS.AI **50%**

arXiv:2607.24097v1 Announce Type: new Abstract: Memory-augmented LLM agents typically answer queries by retrieving relevant memories and feeding them directly to an answer model. This retrieval-as-evidence paradigm assumes retrieved memories are already suitable for reasoning,...

https://arxiv.org/abs/2607.24097

1 **CONF**

Loss-Aware Feature-Map Pruning in Convolutional Neural Networks Using Multi-Armed Bandits

ArXiv CS.AI **50%**

arXiv:2607.22564v1 Announce Type: new Abstract: Convolutional neural networks often contain redundant feature maps that increase storage and inference cost. This paper presents a loss-aware feature-map pruning framework using multi-armed bandits. Feature-map pruning is...

https://arxiv.org/abs/2607.22564

2 **CONF**

Lexical discovery in unknown environments orchestrated by Large Language Models

ArXiv CS.AI **50%**

arXiv:2607.22591v1 Announce Type: new Abstract: Populations of autonomous agents deployed in unknown environments (e.g. planetary or deep-sea exploration) must develop shared vocabularies to refer to entities that have no name in any human language. We propose the Neuro-Symbolic...

https://arxiv.org/abs/2607.22591

3 **CONF**

Beyond a Global Norm: Personalizing Toxicity Sensitivity in Language Models Without Retraining

ArXiv CS.AI **50%**

arXiv:2607.23175v1 Announce Type: cross Abstract: Reducing toxicity is often framed as a global alignment problem, yet perceptions of harmful language are subjective and context-dependent. We present the first comparative evaluation of training-free methods for aligning language...

https://arxiv.org/abs/2607.23175

4 **CONF**

Test-Time Coverage: Test-Conditioned Data Curation for Deployment-Aware Learning

ArXiv CS.AI **50%**

arXiv:2607.22697v1 Announce Type: new Abstract: Deployed AI systems are often trained from broad candidate data pools, necessitating data curation towards the deployment test distribution. However, standard data curation methods score training-side criteria rather than directly...

https://arxiv.org/abs/2607.22697

5 **CONF**

TLA^{+}-Bench: An Execution-Grounded Benchmark and Dataset for Natural-Language to TLA+ Specification Generation

ArXiv CS.AI 50%

arXiv:2607.23425v1 Announce Type: cross Abstract: Large language models increasingly write TLA^{+} formal specifications from natural-language descriptions, but progress is hard to measure: existing resources grade by resemblance to a reference or by whether the output parses,...

https://arxiv.org/abs/2607.23425

6 **CONF**

Aligning Quantum Operators with Large Language Models

ArXiv CS.AI 50%

arXiv:2606.13811v2 Announce Type: replace-cross Abstract: Can Large Language Models (LLMs) understand and reason about quantum operators? Despite their remarkable capabilities in mathematics and symbolic reasoning, LLMs remain inherently blind to quantum representations such as...

https://arxiv.org/abs/2606.13811

7 **CONF**

A scalable online machine learning approach for Stock Recommendation

ArXiv CS.AI 50%

arXiv:2607.23120v1 Announce Type: cross Abstract: Stock recommendation systems face the dual challenge of adapting to rapidly changing market conditions while maintaining low-latency predictions for end users. Traditional batch-trained models fail to capture concept drift, and...

https://arxiv.org/abs/2607.23120

8 **CONF**

Multi-Objective Structured Pruning of LLMs for Latency and Model Size Optimization

ArXiv CS.AI 50%

arXiv:2607.22583v1 Announce Type: new Abstract: Large Language Models (LLMs) have achieved widespread adoption because of their strong reasoning and query-response capabilities. However, deploying them in embedded and edge computing environments remains challenging because of...

https://arxiv.org/abs/2607.22583

9 **CONF**

ParBench: A Benchmark for Reliable Evaluation of LLM Parallel Code Translation

ArXiv CS.AI 50%

arXiv:2607.22588v1 Announce Type: new Abstract: Modern compute-intensive software must migrate across a changing ecosystem of accelerators, programming APIs, compiler stacks, and portability layers, including CUDA, OpenMP, OpenCL, and OpenMP target offload. Large language models...

https://arxiv.org/abs/2607.22588

0 **CONF**

Structure Over Scale: Schema-Constrained Causal Graphs for RAG

ArXiv CS.AI 50%

arXiv:2607.22592v1 Announce Type: new Abstract: Graph-based retrieval-augmented generation (GraphRAG) grounds answers in structured knowledge, but current systems extract entities and relationships exhaustively, producing graphs whose size and construction cost scale with corpus...

https://arxiv.org/abs/2607.22592

1 **CONF**

Opti-Q: A Constraint-Based Optimization Framework for Multi-LLM Question Planning

ArXiv CS.AI **50%**

arXiv:2607.22621v1 Announce Type: new Abstract: While large language models (LLMs) enable strong question answering (QA), budgeted deployment is complicated by nondeterminism and heterogeneous resource profiles (cost, latency, and energy). We present OPTI-Q, a database-inspired,...

https://arxiv.org/abs/2607.22621

2 **CONF**

Differencing the Diffusion Trajectory toward Uncertain Components for Time Series Forecasting

ArXiv CS.AI **50%**

arXiv:2607.22599v1 Announce Type: new Abstract: Diffusion models have become a widely used framework for probabilistic time series forecasting, modeling the distribution of future values given an observed history. In time series forecasting, however, the future continues the...

https://arxiv.org/abs/2607.22599

3 **CONF**

Evaluating Large Language Models for Symbolic Security Protocol Analysis

ArXiv CS.AI **50%**

arXiv:2607.20712v1 Announce Type: cross Abstract: Security protocol verification relies on formal tools such as ProVerif and OFMC. This study evaluates whether Large Language Models (LLMs) can perform comparable analysis. We test GPT and DeepSeek in chat and reasoning modes over...

https://arxiv.org/abs/2607.20712

4 **CONF**

Auditing Alignment Controllability in LLMs via Political Axes

ArXiv CS.AI **50%**

arXiv:2607.23519v1 Announce Type: cross Abstract: Political audits of large language models (LLMs) usually reduce each to one point on a political compass. But that resting point barely matters in deployment: a model must land somewhere, and what counts is how far, and in which...

https://arxiv.org/abs/2607.23519

5 **CONF**

Decentralized Granular Access Control for Agentic AI Systems in Critical Infrastructure

ArXiv CS.AI **50%**

arXiv:2607.22611v1 Announce Type: new Abstract: The deployment of autonomous AI agents in production infrastructure introduces fundamental security challenges that traditional role-based access control (RBAC) models cannot address. Unlike deterministic automation, AI agents...

https://arxiv.org/abs/2607.22611

6 **CONF**

Unified Embodied VLM Reasoning with Robotic Action via Autoregressive Discretized Pre-training

ArXiv CS.AI **50%**

arXiv:2512.24125v3 Announce Type: replace-cross Abstract: General-purpose robotic systems operating in open-world environments must achieve both broad generalization and high-precision action execution, a combination that remains challenging for existing Vision-Language-Action...

https://arxiv.org/abs/2512.24125

7 **CONF**

TokenMem: Faithful Knowledge Injection for Frozen LLMs

ArXiv CS.AI **50%**

arXiv:2607.22625v1 Announce Type: new Abstract: Retrieval-augmented generation (RAG) enhances large language models (LLMs) with external knowledge, but suffers from knowledge conflicts: when retrieved information contradicts parametric memory, the shared self-attention pathway...

https://arxiv.org/abs/2607.22625

8 **CONF**

PRESTO: Prefix-Aligned Tree Drafting for Diffusion Speculative Decoding

ArXiv CS.AI **50%**

arXiv:2607.22634v1 Announce Type: new Abstract: Diffusion Large Language Models (dLLMs) have emerged as a promising alternative to autoregressive (AR) LLMs, generating tokens in parallel. This makes them effective draft models for speculative decoding (SD), producing an entire...

https://arxiv.org/abs/2607.22634

9 **CONF**

TRACE: Business Rule-Grounded Reasoning Curriculum for Knowledge-Preserving Parametric Tool Retrieval in Enterprise LLMs

ArXiv CS.AI **50%**

arXiv:2607.22639v1 Announce Type: new Abstract: Parametric retrieval enables LLMs to retrieve tools implicitly by assigning each API a unique virtual token and training the model to generate it via constrained beam search. Toolsense shows that this regime has two critical...

https://arxiv.org/abs/2607.22639

0 **CONF**

Reason Before You Retrieve: Agentic Planning for Multi-modal RAG

ArXiv CS.AI **50%**

arXiv:2607.22643v1 Announce Type: new Abstract: Multimodal retrieval-augmented generation (mRAG) aims to answer image-text queries with external knowledge, but most existing systems still retrieve directly from raw multimodal input over a flat evidence space. This design often...

https://arxiv.org/abs/2607.22643

1 **CONF**

SkillSieve: A Hierarchical Triage Framework for Detecting Malicious AI Agent Skills

ArXiv CS.AI **50%**

arXiv:2604.06550v3 Announce Type: replace-cross Abstract: Agent skills combine natural-language instructions with executable code while inheriting an agent's filesystem, credential, and network access. Attacks can span prose and files, whereas regex and code-only analyzers cover...

https://arxiv.org/abs/2604.06550

2 **CONF**

Extracting Algorithms in Pre-trained LLMs: A Case on Hidden Markov Models

ArXiv CS.AI **50%**

arXiv:2607.22646v1 Announce Type: new Abstract: Large language models (LLMs) display a striking ability to predict next observations from Hidden Markov Models (HMMs) via in-context learning (ICL), but the algorithm underlying this capability remains undetermined: prior work has...

https://arxiv.org/abs/2607.22646

3 **CONF**

PTStore (Prefix Tensor Store): Distributed Prefix Caching and Replication for High Throughput Inference Serving

ArXiv CS.AI 50%

arXiv:2607.22648v1 Announce Type: new Abstract: Inspired by the design of client caching in Content Delivery Networks (CDNs), PTStore distributes and replicates popular tensors that form reusable KV cache prefixes, which are the main technique used by state of art approaches to...

https://arxiv.org/abs/2607.22648

4 **CONF**

STAIF: A Stage-wise Optimization for Complex Instruction Following

ArXiv CS.AI 50%

arXiv:2607.22649v1 Announce Type: new Abstract: Following complex instructions with multiple explicit constraints remains a fundamental challenge for large language models (LLMs). Existing alignment methods, such as DPO, optimize holistic reward signals that often underemphasize...

https://arxiv.org/abs/2607.22649

5 **CONF**

ARdena: Scenario-driven control of real-time LLM agents

ArXiv CS.AI 50%

arXiv:2607.22651v1 Announce Type: new Abstract: Large language models (LLMs) have enabled increasingly capable conversational agents, but reliably controlling their behavior in real-time interactive environments remains a significant challenge. Existing approaches often rely on...

https://arxiv.org/abs/2607.22651

6 **CONF**

KG2Code: Bridging Knowledge Graphs and Large Language Models via Executable Code for Question Answering

ArXiv CS.AI 50%

arXiv:2607.22652v1 Announce Type: new Abstract: Recent research has explored the integration of knowledge graphs (KGs) with large language models (LLMs) to enhance their performance on downstream knowledge-intensive tasks, particularly knowledge graph question answering (KGQA)....

https://arxiv.org/abs/2607.22652

7 **CONF**

Do Language Models Converge to Themselves? Recursive Self-Refinement as Textual Relaxation

ArXiv CS.AI 50%

arXiv:2607.22653v1 Announce Type: new Abstract: Large language models are increasingly used in recursive refinement workflows, where an initial draft is repeatedly revised by the same model. Despite their growing use, the long-term dynamics of such workflows remain poorly...

https://arxiv.org/abs/2607.22653

8 **CONF**

EventOD: Event-Aware OD Flow Generation via LLM-Guided Semantic Modulation

ArXiv CS.AI 50%

arXiv:2607.22655v1 Announce Type: new Abstract: Estimating origin-destination (OD) flows under disruptive events is important for disaster response and urban resilience. Existing deep OD models trained on routine mobility often degrade when extreme events abruptly alter regional...

https://arxiv.org/abs/2607.22655

9 CONF

CuraWeb: Joint Optimization of Quality, Redundancy, and Diversity for Web-Scale Pretraining Data

ArXiv CS.AI **50%**

arXiv:2607.22662v1 Announce Type: new Abstract: Open-web corpora curated via highly selective filters, such as FineWeb-Edu and DCLM, constitute the core of LLM pretraining data and have significantly advanced LLM performance. However, these pipelines typically rely on singular...

https://arxiv.org/abs/2607.22662

0 CONF

A Coulomb Particle Model for Learning Kernel Attention in Transformers

ArXiv CS.AI **50%**

arXiv:2607.23869v1 Announce Type: cross Abstract: Randomized features provide a scalable approximation to kernel machines, but their performance depends strongly on the choice of feature distribution. We propose a particle-based method that learns this distribution by optimizing...

https://arxiv.org/abs/2607.23869

1 CONF

How LLM Task-Adaptation Reshapes Alignment: A Multi-dimensional Study of Behavioral and Representational Drift

ArXiv CS.AI **50%**

arXiv:2607.22676v1 Announce Type: new Abstract: Post-training is a key mechanism for adapting large language models to downstream tasks. While prior work suggests that task adaptation can alter a model's pre-existing alignment, especially its safety behavior, its broader effects...

https://arxiv.org/abs/2607.22676

2 CONF

LLM-SoccerArena: Benchmarking LLMs on Real-World Predictions in Sports

ArXiv CS.AI **50%**

arXiv:2607.24573v1 Announce Type: new Abstract: Large language models (LLMs) increasingly support decisions about uncertain future events, yet evaluating their ability to forecast real-world outcomes remains difficult. In particular, existing benchmarks are typically static and...

https://arxiv.org/abs/2607.24573

3 CONF

HiLLTS: Zero-Shot Hierarchical LLM-Guided Traffic Signal Control for Sustainable Transportation

ArXiv CS.AI **50%**

arXiv:2607.22691v1 Announce Type: new Abstract: Urban traffic congestion significantly increases fuel consumption, greenhouse gas emissions, and commuter delays, resulting in substantial economic losses and environmental harm in modern cities. Traditional traffic signal control...

https://arxiv.org/abs/2607.22691

4 CONF

Risk Governance for Generative AI Mental Health Support: A Multi-Turn Safety Architecture

ArXiv CS.AI **50%**

arXiv:2607.22692v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used for emotional support despite lacking mechanisms to safely govern evolving mental health risk. Existing safety approaches primarily detect risk but rarely shape how models respond...

https://arxiv.org/abs/2607.22692

5 CONF

Bayesian Repetition Penalty: A Principled Adjacent-Conditional Framework for Reversing Attention Collapse in Autoregressive Language Models

ArXiv CS.AI 50%

arXiv:2607.22694v1 Announce Type: new Abstract: Attention collapse in autoregressive language models -- manifested as repetitive token loops where the model becomes trapped in self-reinforcing attractors -- is a persistent pathology that existing decoding-time heuristics fail to...

https://arxiv.org/abs/2607.22694

6 CONF

PANOPTICON: A PII-Based Assemblage of Naturalistic Output Tokens for Investigating Privacy Leakage Within LLM Context Window

ArXiv CS.AI 50%

arXiv:2607.22695v1 Announce Type: new Abstract: Large Language Models (LLMs) are capable of generalizing human language for the completion of never-before-seen tasks, leading to widespread deployment. While this automation provides clear utility, completing these tasks often...

https://arxiv.org/abs/2607.22695

7 CONF

Quality Action Assurance: Multimodal Verification of Examiner Claims in VR OSCEs

ArXiv CS.AI 50%

arXiv:2607.19063v2 Announce Type: replace Abstract: Objective Structured Clinical Examinations (OSCEs) are the gold standard for assessing clinical competence, yet scoring remains vulnerable to examiner subjectivity, fatigue, and cognitive bias. Standard examiner validation via...

https://arxiv.org/abs/2607.19063

8 CONF

SyRuP: Enhancing System-Prompt Following via Reward-Guided Prediction in LLM Decoding

ArXiv CS.AI 50%

arXiv:2607.23991v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly controlled through system prompts that specify roles, styles, formats, and safety requirements. However, models follow these prompts only implicitly through in-context learning, which...

https://arxiv.org/abs/2607.23991

9 CONF

Spatial Reasoning in LLM Game Agents: Impact of Causal Context and Multi-Step Planning

ArXiv CS.AI 50%

arXiv:2607.22732v1 Announce Type: new Abstract: LLM-based game agents often perform poorly on more complex tasks. This work examines whether these failures are linked to limited spatial reasoning and evaluates whether causal prompt augmentation and multi-step planning can...

https://arxiv.org/abs/2607.22732

0 CONF

Physical AI Governance: From Theory to Practice Across Life Cycle

ArXiv CS.AI 50%

arXiv:2607.22877v1 Announce Type: new Abstract: With the emergence of Physical AI, artificial intelligence is extending beyond screen-based applications to embodied systems that perceive, interact with, and act in the physical world. Unlike traditional AI, Physical AI operates...

https://arxiv.org/abs/2607.22877

1 CONF

How Well Can AI Generate Backlogs from App Mockups?

ArXiv CS.AI 50%

arXiv:2607.22902v1 Announce Type: new Abstract: Creating sprint backlogs requires considerable effort, as items such as epics, user stories, and tasks can be missed or inconsistently specified. We propose a multimodal approach to support backlog generation from visual app...

https://arxiv.org/abs/2607.22902

2 CONF

SAGE: Safety-First Defense-in-Depth Guardrails for Verified Lifecycle Control of High-Impact Generative AI

ArXiv CS.AI 50%

arXiv:2607.22926v1 Announce Type: new Abstract: High-impact generative AI makes catastrophic misuse a lifecycle-control problem, not merely a prompt-filtering problem. SAGE is a safety-first, authorization-separated architecture in which credible catastrophic-enablement risk...

https://arxiv.org/abs/2607.22926

3 CONF

Share No More Than the Request Requires: Federated Disclosure for Perspective-Aware AI

ArXiv CS.AI 50%

arXiv:2607.22953v1 Announce Type: new Abstract: Modern AI systems bring societal risks such as mass surveillance, extreme concentrations of power, and loss of user autonomy---calling into question a model where third-parties collect and control massive amounts of user data....

https://arxiv.org/abs/2607.22953

4 CONF

ConsistencyGate: Preventing Memory Contamination in LLM Agents via Self-Consistency Admission Control

ArXiv CS.AI 50%

arXiv:2607.22962v1 Announce Type: new Abstract: LLM agents that operate over many turns accumulate facts in an external memory store and reuse them as premises for downstream reasoning. A hallucinated fact written at one step therefore persists as a false premise for every...

https://arxiv.org/abs/2607.22962

5 CONF

Stress-testing large language model agents in a robotic chemistry laboratory

ArXiv CS.AI 50%

arXiv:2607.23045v1 Announce Type: new Abstract: AI is evaluated through knowledge, reasoning and plan generation, yet scientific agency requires reliable physical action and adaptation to evidence. Here, we use a robotic chemistry laboratory as a physical-world testbed to make...

https://arxiv.org/abs/2607.23045

6 CONF

CachedSearch: Training-Free Cached Exploration for Test-Time Search in Video Diffusion

ArXiv CS.AI 50%

arXiv:2607.23159v1 Announce Type: new Abstract: Test-time search lets small video diffusion models rival larger ones, but costs 2-10x more. All candidates are fully denoised, although most are discarded. Training-free caching makes each rollout 2-3x faster at near-lossless...

https://arxiv.org/abs/2607.23159

7 CONF

An Ontology for Machine Learning Interatomic Potentials

ArXiv CS.AI 50%

arXiv:2607.23219v1 Announce Type: new Abstract: Machine learning interatomic potentials (MLIPs) approximate quantum-mechanical energies and forces---conventionally computed by density functional theory (DFT) or wave-function methods---at a fraction of the cost. The field...

https://arxiv.org/abs/2607.23219

8 CONF

CAPT: A Multi-task Continuous Autoregressive Transformer enabling Cross-dataset and Cross-species Transfer for Calcium Population Dynamics

ArXiv CS.AI 50%

arXiv:2607.23258v1 Announce Type: new Abstract: Large-scale calcium imaging has created an opportunity to build foundation-style models for neural population dynamics, but a central question remains unresolved: \textbf{whether a model pretrained on one collection of recordings...

https://arxiv.org/abs/2607.23258

9 CONF

TopoFE: topology-aware LLM-guided Automated Feature Engineering

ArXiv CS.AI 50%

arXiv:2607.23286v1 Announce Type: new Abstract: Automatic feature engineering (AutoFE) for tabular learning can be naturally formulated as a program synthesis problem, where the objective is to discover predictive feature transformations from an exponentially large search space....

https://arxiv.org/abs/2607.23286

0 CONF

SIREN: Towards End-to-End Extreme-Weather Early Warning with Experience-Grounded LLM Agents

ArXiv CS.AI 50%

arXiv:2607.24588v1 Announce Type: new Abstract: Early warning of extreme weather is essential for mitigating the societal, economic, and environmental risks posed by hazardous weather events. However, expert-centered warning workflows are costly, labor-intensive, and difficult...

https://arxiv.org/abs/2607.24588

1 CONF

E-Bench: Benchmarking Multi-Step Tool-Use Agents in Real-World Product Scenarios

ArXiv CS.AI 50%

arXiv:2607.23722v1 Announce Type: new Abstract: Large Language Models (LLMs) are increasingly deployed as agents that interact with stateful environments over multiple steps: gathering hidden information, composing tool calls, and committing state changes. We refer to this...

https://arxiv.org/abs/2607.23722

2 CONF

Confidently Wrong: Exception Chain Collapse in Frontier LLM Rule Evaluation

ArXiv CS.AI 50%

arXiv:2607.23386v1 Announce Type: new Abstract: We document a failure class in frontier large language models -- exception chain collapse -- observed in eligibility evaluation under nested conditional rules of the form "A is required UNLESS B applies, UNLESS C overrides B". The...

https://arxiv.org/abs/2607.23386

3 CONF

Inference-Time Consensus for Mitigating Hidden Behaviors from LLM Fine-Tuning

ArXiv CS.AI 50%

arXiv:2607.23394v1 Announce Type: new Abstract: Recent work shows that fine-tuning language models on even a small amount of poisoned data can install targeted misbehavior, and ostensibly benign data can transmit hidden preferences that generalize broadly. Standard defenses,...

https://arxiv.org/abs/2607.23394

4 CONF

Do LLMs Know Their Vulnerable Scenarios?

ArXiv CS.AI 50%

arXiv:2607.23496v1 Announce Type: new Abstract: Safety-aligned large language models are trained to refuse harmful requests, yet embedding the same requests in particular scenarios can bypass their safeguards. Existing red-teaming methods empirically identify effective scenarios...

https://arxiv.org/abs/2607.23496

5 CONF

ObsDriveBench: Benchmarking Multimodal Understanding under Adverse Weather with Observability Awareness

ArXiv CS.AI 50%

arXiv:2607.23537v1 Announce Type: new Abstract: Autonomous driving under adverse weather remains a critical challenge, yet existing vision-language benchmarks mainly evaluate under standard conditions, synthetic corruptions, or single modality. As a result, it remains unclear...

https://arxiv.org/abs/2607.23537

6 CONF

Token-Region Guided Cross-Attention Fusion for Multimodal Affect Interpretation

ArXiv CS.AI 50%

arXiv:2607.23493v1 Announce Type: cross Abstract: Automated analysis of multimodal content on social networks has become a critical task for understanding public sentiment and information diffusion in the digital age. However, classifying internet memes remains computationally...

https://arxiv.org/abs/2607.23493

7 CONF

Expert Behavior Prior Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.21302v2 Announce Type: replace Abstract: Behavior prior reinforcement learning (BPRL) has emerged as a promising paradigm to improve sample efficiency in online reinforcement learning (RL) by leveraging policy priors derived from offline demonstrations. However, most...

https://arxiv.org/abs/2607.21302

8 CONF

Offline-to-Online Creative Optimization with Generative Models and Adaptive Testing

ArXiv CS.AI 50%

arXiv:2607.23696v1 Announce Type: new Abstract: Ad creative optimization is increasingly constrained by evaluation rather than generation. Generative models can produce many plausible creatives, but reliable evaluation requires online experiments, in which only a limited slate...

https://arxiv.org/abs/2607.23696

9 CONF

From RLVR to RLSVR: Task Transformation Induces Self-Verifiable Rewards for Open-Ended LLM Self-Improvement

ArXiv CS.AI 50%

arXiv:2607.23802v1 Announce Type: new Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) has driven recent progress in reasoning-oriented large language models (LLMs) by enabling large-scale optimization. However, its applicability remains largely limited to domains...

https://arxiv.org/abs/2607.23802

0 CONF

Do Visual Features Improve Other-Initiated Repair Detection? A Dyadic Multimodal Approach

ArXiv CS.AI 50%

arXiv:2607.23845v1 Announce Type: new Abstract: Other-initiated Self-repair, or in short Other-initiated Repair (OIR), is an essential mechanism in conversational interaction, whereby a recipient signals a problem in speaking, hearing, or understanding, prompting the previous...

https://arxiv.org/abs/2607.23845

1 CONF

GOTS: Greedy Orthogonal Token Selection for High-Resolution Vision-Language Models

ArXiv CS.AI 50%

arXiv:2607.23913v1 Announce Type: new Abstract: Modern vision-language models (VLMs) increasingly rely on dynamic or high-resolution visual encoding, producing thousands of visual tokens that substantially increase downstream language-model inference cost. Existing...

https://arxiv.org/abs/2607.23913

2 CONF

Plato-Bio: verification-first biological novelty screening with temporal rediscovery and structural benchmarks

ArXiv CS.AI 50%

arXiv:2607.23975v1 Announce Type: new Abstract: Large language model research agents can connect literature retrieval, analysis code, and manuscript preparation, but coherent output does not establish scientific validity. We developed Plato-Bio, a biology-routed extension of the...

https://arxiv.org/abs/2607.23975

3 CONF

A Motion-Aware Vector Quantization Framework with Centroid Reuse for Efficient VLA Inference

ArXiv CS.AI 50%

arXiv:2607.24148v1 Announce Type: new Abstract: Vision-Language-Action (VLA) models have demonstrated strong potential for embodied AI, yet their high inference latency on GPUs limits real-time deployment. Existing accelerators, such as Dadu-Corki, improve efficiency but treat...

https://arxiv.org/abs/2607.24148

4 CONF

The Half-Lives of Generative-AI Evidence: A 40-Record Audit, a Claim-Currency Framework, and a Reflexive Case of Frontier-Model-Assisted Research

ArXiv CS.AI 50%

arXiv:2607.24032v1 Announce Type: new Abstract: Generative-AI evaluations can become historical before publication, yet calendar age does not affect every conclusion equally. This paper has two linked purposes. First, it audits a maximum-variation purposive corpus of 40...

https://arxiv.org/abs/2607.24032

5 CONF

The Cost of Knowing: A Resource-Aware Protocol for Benchmarking Hallucination Beyond Static Leaderboards

ArXiv CS.AI 50%

arXiv:2607.24063v1 Announce Type: new Abstract: On standard factuality tasks, frontier models now cluster near the top of the scale. The question is therefore shifting from how factual a system is toward how much compute that factuality costs. Static leaderboards score...

https://arxiv.org/abs/2607.24063

6 CONF

Grading the Narrators: An Isnad-Rijal Framework for Claim-Level Provenance in Multi-Agent Knowledge Systems

ArXiv CS.AI 50%

arXiv:2607.24117v1 Announce Type: new Abstract: Modern multi-agent knowledge systems increasingly accumulate knowledge through chains of autonomous transformations rather than direct retrieval. Existing provenance work records what happened - execution traces, tool calls,...

https://arxiv.org/abs/2607.24117

7 CONF

Speed Reading Tool Powered by Artificial Intelligence for Students with ADHD, Dyslexia, and Short Attention Span

ArXiv CS.AI 50%

arXiv:2307.14544v2 Announce Type: replace-cross Abstract: This paper presents an artificial intelligence tool designed to assist students with dyslexia, ADHD, and short attention spans in processing text-based information more efficiently. The proposed solution addresses both...

https://arxiv.org/abs/2307.14544

8 CONF

Integrating Factual and Normative Industrial Knowledge via Constraint-Aware Graph Attention for Process Plan Recommendation

ArXiv CS.AI 50%

arXiv:2607.24213v1 Announce Type: new Abstract: Integrating heterogeneous industrial knowledge, including factual relations and decision constraints, remains a core challenge in industrial information systems. Machining process planning exemplifies this problem because engineers...

https://arxiv.org/abs/2607.24213

9 CONF

Generative Artificial Intelligence (GenAI) to convert images of queuing networks into verifiable simulation models: an open-weight LLM workflow approach

ArXiv CS.AI 50%

arXiv:2607.24259v1 Announce Type: new Abstract: Recent work has explored the use of Large Language Models (LLMs) to automate simulation model building, typically by generating executable code directly from natural language descriptions. However, this raises challenges for...

https://arxiv.org/abs/2607.24259

00 CONF

Gubernaut: A Deterministic Homeostatic Controller for Affect-Regulated LLM Agents, Validated Across Independent Model Families

ArXiv CS.AI 50%

arXiv:2607.24339v1 Announce Type: new Abstract: Large language model (LLM) agents inherit reactive failure modes: escalation under provocation, sycophantic drift under flattery, perseveration when stuck. These are failures of propensity, not capability; they concern what a model...

https://arxiv.org/abs/2607.24339

01 CONF

Simulating Tenant Responses to Energy Policy Interventions with Transaction-Cost-Aware LLM Age

ArXiv CS.AI 50%

arXiv:2607.24341v1 Announce Type: new Abstract: Recent studies use Large language models (LLMs) to simulate human opinions and decisions by prompting models with demographic, attitudinal, or persona-based descriptions. Yet such simulations rarely model the practical, cognitive,...

https://arxiv.org/abs/2607.24341

02 CONF

Failures Reveal What Metrics Miss: An Evidence-Driven Agent for Recursive Refinement of ECG Classifiers

ArXiv CS.AI 50%

arXiv:2607.24419v1 Announce Type: new Abstract: Deep models have substantially advanced 12-lead ECG classification, yet their refinement still relies heavily on human experts to inspect failures and iteratively revise classifier designs. Recent LLM-based agents have demonstrated...

https://arxiv.org/abs/2607.24419

03 CONF

Reason-Mediated Behavioral Models for Auditing LLM Social Simulators

ArXiv CS.AI 50%

arXiv:2607.24649v1 Announce Type: new Abstract: Large language models are increasingly used as social simulators, including as synthetic survey respondents. Most evaluations ask whether simulated outcomes resemble human outcomes. We argue that this is necessary but too weak: a...

https://arxiv.org/abs/2607.24649

04 CONF

LLM-Assisted Ontology Engineering and Construction of a French Legal Knowledge Graph

ArXiv CS.AI 50%

arXiv:2607.24551v1 Announce Type: new Abstract: Maintenance regulations are complex legal texts that are difficult to exploit when addressing a specific case and challenging to integrate into operational systems. This paper presents a two-stage LLM-assisted workflow for French...

https://arxiv.org/abs/2607.24551

05 CONF

Artificial Intelligence and Innovation Ecosystem: Evolutionary Developments, Challenges, and Future Directions

ArXiv CS.AI 50%

arXiv:2607.24589v1 Announce Type: new Abstract: The development of the Innovative Ecosystem (IE) presents a new paradigm for economic integration, collaborative advancement, and shared achievements. The rise of Artificial Intelligence (AI) has significantly accelerated the...

https://arxiv.org/abs/2607.24589

06 CONF

ERUnderstand: Evaluating Vision-Language Models on Structured ER Diagrams

ArXiv CS.AI 50%

arXiv:2607.24707v1 Announce Type: new Abstract: Entity-Relationship Diagrams (ERDs) are central to conceptual database design, yet they are typically available only as rendered images rather than machine-readable schemas, limiting AI-assisted database engineering. We introduce...

https://arxiv.org/abs/2607.24707

07 CONF

Semalith v1.4: A Calibrated 184M Safety Classifier Achieving State-of-the-Art Prompt-Injection Detection at 44x Fewer Parameters than Llama-Guard-3-8B

ArXiv CS.AI 50%

arXiv:2607.22545v1 Announce Type: cross Abstract: Deploying large language models in financial-services and agentic settings requires safety classifiers that simultaneously handle prompt injection, regulatory compliance, and general harm, a combination no existing open guardrail...

https://arxiv.org/abs/2607.22545

08 CONF

Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests

ArXiv CS.AI 50%

arXiv:2607.22883v1 Announce Type: cross Abstract: While Large Language Models (LLMs) show great promise for automating unit test generation, recent studies suggest that the quality of generated tests can be negatively impacted when models are prompted with buggy code. This paper...

https://arxiv.org/abs/2607.22883

09 CONF

Revitalizing Public Urban Places through Cultural and Political Memory: A Technological Approach with LLMs and Augmented Reality

ArXiv CS.AI 50%

arXiv:2607.22613v1 Announce Type: cross Abstract: This paper explores the intersection of memory, place, and identity, examining how new technologies, particularly Apple Vision Pro, can illuminate this nexus. Leveraging digital twins and virtual reality, it investigates how...

https://arxiv.org/abs/2607.22613

10 CONF

AI-Assisted Causal Inference and Mediation Analyses of Environmental and Psychosocial Determinants of Subjective Cognitive Difficulties in the All of Us Research Program

ArXiv CS.AI 50%

arXiv:2607.22640v1 Announce Type: cross Abstract: Short-term environmental exposures have been linked to cognitive and behavioral outcomes, although many reported associations may reflect broader geographic and contextual differences. Using longitudinal data from the All of Us...

https://arxiv.org/abs/2607.22640

11 CONF

SetGo: Metadata Readiness for Scientific AI Datasets

ArXiv CS.AI 50%

arXiv:2607.22677v1 Announce Type: cross Abstract: Scientific datasets intended for AI use require both computational readiness for model training and metadata readiness for discovery, sharing, and reuse. The Readiness Engine for Data Integration (REDI) addresses computational...

https://arxiv.org/abs/2607.22677

12 CONF

Towards Nexus-Score: Metadata Gaps Limit Scholarly AI Attribution

ArXiv CS.AI 50%

arXiv:2607.22684v1 Announce Type: cross Abstract: Artificial intelligence systems increasingly mediate how science is found and credited. We asked whether missing metadata prevents AI systems from crediting work. As a boundary test, an AI system citing without access to...

https://arxiv.org/abs/2607.22684

13 CONF

MegaSlide-DiT: Memory-Centric Adaptation and Deformable Local Attention for Efficient Video Diffusion

ArXiv CS.AI 50%

arXiv:2607.22696v1 Announce Type: cross Abstract: High-resolution video diffusion models built on Diffusion Transformers (DiTs) deliver strong fidelity but quickly exhaust the memory budget of a single workstation. A 100 billion-plus parameter DiT easily requires over a terabyte...

https://arxiv.org/abs/2607.22696

14 CONF

Visible-Light Imaging Diagnosis of Neutral Particle Emission Tomography in the Tokamak Divertor: An Efficient Transformer-based Surrogate Model

ArXiv CS.AI 50%

arXiv:2607.22704v1 Announce Type: cross Abstract: Nuclear fusion has made significant progress in recent years and is expected to become one of the most important pathways to addressing global energy challenges. This paper focuses on observing plasma using visible-light cameras,...

https://arxiv.org/abs/2607.22704

15 CONF

RMS@CC-MMD 2026: Multimodal Misogyny Detection via Geometric Interaction and Multi-View Consensus

ArXiv CS.AI 50%

arXiv:2607.22709v1 Announce Type: cross Abstract: The proliferation of internet memes has introduced new complexities to automated content moderation, particularly in detecting misogyny. Memes often rely on a semantic clash between visual and textual modalities, where hateful...

https://arxiv.org/abs/2607.22709

16 CONF

Structural Preservation Governs Data Augmentation in Deep Learning-Based Laser Speckle Material Classification

ArXiv CS.AI 50%

arXiv:2607.22725v1 Announce Type: cross Abstract: Data augmentation is routinely used to improve generalization in image classification, but the assumptions underlying standard policies are poorly matched to coherent imaging. Laser speckle patterns are not generic textures; they...

https://arxiv.org/abs/2607.22725

17 CONF

AI-generated Images Challenge Visual Trust in High-risk Scenarios

ArXiv CS.AI 50%

arXiv:2607.22745v1 Announce Type: cross Abstract: Rapid advances in image generation are eroding the evidentiary value of visual content in settings where authenticity can affect public safety and personal reputation. Yet existing detection benchmarks rarely examine synthetic...

https://arxiv.org/abs/2607.22745

18 CONF

Spectral Dynamics of Semantic Drift in Clinical Multi-Agent Language Model Networks

ArXiv CS.AI 50%

arXiv:2607.22758v1 Announce Type: cross Abstract: The integration of iterative LLMs within multi-agent diagnostic frameworks requires a rigorous quantitative reevaluation of underlying communication topologies. Frequently used architectural paradigms depend on scale-free or...

https://arxiv.org/abs/2607.22758

19 CONF

Beyond Shapley: An Influence-Based Data Auditing Pipeline for LLM Alignment and Evaluation

ArXiv CS.AI 50%

arXiv:2607.22766v1 Announce Type: cross Abstract: The alignment of Large Language Models (LLMs) is increasingly bottlenecked by data quality. As datasets scale, massive preference and instruction-tuning corpora inevitably accumulate hidden structural contradictions, safety...

https://arxiv.org/abs/2607.22766

20 CONF

Multimodal Surface EMG Hand Gesture Recognition Using Query-Based Transformers for Prosthetic Control

ArXiv CS.AI 50%

arXiv:2607.22779v1 Announce Type: cross Abstract: Hand gesture recognition via surface electromyography (sEMG) is fundamental to prosthetic control. In this field, deep learning approaches have become the gold standard. However, current architectures struggle to scale; model...

https://arxiv.org/abs/2607.22779

21 CONF

Optimizing Transformer Neural Network for Real-Time Outlier Detection on FPGAs

ArXiv CS.AI 50%

arXiv:2607.22786v1 Announce Type: cross Abstract: In this work, we explore how the inference time of a Transformer Neural Network can be efficiently optimized with applications to real-time anomaly detection in financial time series. The financial time series are price series...

https://arxiv.org/abs/2607.22786

22 CONF

FMOPF: Latent Flow Matching with Constraint-Aware Interaction Priors for AC Optimal Power Flow

ArXiv CS.AI 50%

arXiv:2607.22788v1 Announce Type: cross Abstract: AC optimal power flow determines the minimum-cost generation dispatch under nonlinear power balance constraints and is solved thousands of times daily in electricity market operations. Learning a direct mapping from load...

https://arxiv.org/abs/2607.22788

23 CONF

Multimodal Domain Generalization for Depression Detection: An Attention-Based BiLSTM Network with Domain-Adversarial Training

ArXiv CS.AI 50%

arXiv:2607.22794v1 Announce Type: cross Abstract: Automatic depression detection with deep learning has shown promise but often suffers from limited generalization due to domain shift arising from inter-speaker variability. To address this critical issue, we present the first...

https://arxiv.org/abs/2607.22794

24 CONF

DeepFaith: Evidence-Grounded LLMs for Faithful Incident Reporting in Multi-Stage APT Defense

ArXiv CS.AI 50%

arXiv:2607.24348v1 Announce Type: cross Abstract: Advanced Persistent Threats (APTs) are difficult to detect and interpret due to their multi-stage and stealthy nature. While recent autonomous defense systems leverage provenance graphs and learning-based models for detection and...

https://arxiv.org/abs/2607.24348

25 CONF

LithoFormer: A Robust Framework for Stratigraphic Inference via Transformers

ArXiv CS.AI 50%

arXiv:2607.22804v1 Announce Type: cross Abstract: Accurate geological characterization of subsurface reservoirs from well log data is essential to support projects such as carbon capture and storage (CCS), geothermal development, and extraction of natural resources. Existing...

https://arxiv.org/abs/2607.22804

26 CONF

From Hybrid Mechanistic--Data-Driven Modeling Toward Neuro-Symbolic AI: What, Why, and How

ArXiv CS.AI 50%

arXiv:2607.22811v1 Announce Type: cross Abstract: Hybrid mechanistic/data-driven models, which combine first-principles with learned components, are increasingly used in process engineering and scientific machine learning. Common hybrid modeling designs are specified primarily...

https://arxiv.org/abs/2607.22811

27 CONF

Language-Routed RAG and Direct Option Scoring for Multilingual Financial QA: DS@GT at FinMMEval

ArXiv CS.AI 50%

arXiv:2607.22841v1 Announce Type: cross Abstract: We present DS@GT's submission to FinMMEval 2026 Task 1, a multilingual financial exam question answering benchmark spanning English, Spanish, Greek, Chinese, and Hindi. Financial certification exams such as the CFA, EFPA, and CPA...

https://arxiv.org/abs/2607.22841

28 CONF

Robustifying pathology foundation models via fine-tuning

ArXiv CS.AI 50%

arXiv:2607.22861v1 Announce Type: cross Abstract: Pathology foundation models (FMs) produce powerful tile-level representations which remain sensitive to scanner and staining variability, undermining deployment across laboratories. We develop a novel fine-tuning recipe that...

https://arxiv.org/abs/2607.22861

29 CONF

Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)

ArXiv CS.AI 50%

arXiv:2607.22880v1 Announce Type: cross Abstract: Recent advances in large language models (LLMs) have driven growing interest in using LLMs to automate test generation. Prior work commonly evaluates generated test suites using proxy metrics such as code coverage and mutation...

https://arxiv.org/abs/2607.22880

30 CONF

Choosing a Text Embedding Model: A Practical Benchmarking and Decision Framework

ArXiv CS.AI 50%

arXiv:2607.23507v1 Announce Type: cross Abstract: Choosing the right text embedding model is one of the most consequential -- and most frequently under-examined -- decisions in building a retrieval or search system, yet the model that tops a leaderboard is rarely the best choice...

https://arxiv.org/abs/2607.23507

31 CONF

Traceable LLM Reasoning for Fake-Order Fraud Detection

ArXiv CS.AI 50%

arXiv:2607.23075v1 Announce Type: cross Abstract: Detecting fake-order fraud at scale remains a critical challenge for large online-to-offline (O2O) service platforms, as existing approaches often rely on expert-designed features, produce black-box decisions, and provide limited...

https://arxiv.org/abs/2607.23075

32 CONF

Building AI That Works: ESnet's Pragmatic Approach to AI-Driven Operational Excellence

ArXiv CS.AI 50%

arXiv:2607.22948v1 Announce Type: cross Abstract: The ORBIT (Operations Responses and Business Intelligence Toolkit) project was initiated to assess agentic AI for the upcoming ESnet 7 initiative and to address persistent operational pain points in the Network Operations Center...

https://arxiv.org/abs/2607.22948

33 CONF

FilmBench: A Film-Grade Benchmark for Cinematic Video Generation

ArXiv CS.AI 50%

arXiv:2607.24241v1 Announce Type: cross Abstract: Progress in video generation keeps narrowing the visual gap between AI-generated and professionally produced footage, yet most benchmarks still draw prompts from web sources or LLM templates and score them with untrained, generic...

https://arxiv.org/abs/2607.24241

34 CONF

Label-free Industrial Fault Detection via Adversarial Inverse Reinforcement Learning: A System for Run-to-Failure Prognostics

ArXiv CS.AI 50%

arXiv:2607.22987v1 Announce Type: cross Abstract: Machinery fault detection (MFD) remains heavily reliant on supervised learning, which struggles with the scarcity of fault labels in real-world settings. While reinforcement learning (RL) offers a framework to model the...

https://arxiv.org/abs/2607.22987

35 CONF

Adversarial Test-Hardening for AI-Written Code: An Instrument Autopsy and a Pre-Registered Causal Estimate of the Critic Loop

ArXiv CS.AI 50%

arXiv:2607.23002v1 Announce Type: cross Abstract: Large language models increasingly write both code and the tests meant to check it; coverage records what ran, not what was verified. We study an adversarial test-hardening loop under a mechanical oracle: a Tester model writes...

https://arxiv.org/abs/2607.23002

36 CONF

All in One: Generative Modeling as Mean-Field Game Design

ArXiv CS.AI 50%

arXiv:2607.23026v1 Announce Type: cross Abstract: Mean-field games (MFGs) offer a unifying lens on continuous-time generative modeling: a cost tuple recovering twelve prominent models---Continuous Normalizing Flows, OT-Flow, Score-based Models, Schr\"{o}dinger Bridges, and...

https://arxiv.org/abs/2607.23026

37 CONF

Through the Bottleneck: How Multi-head Latent Attention Separates Content from Position in Language Models

ArXiv CS.AI 50%

arXiv:2607.23054v1 Announce Type: cross Abstract: Multi-head Latent Attention (MLA), introduced in DeepSeek-V2, compresses key-value pairs through a shared low-rank bottleneck (cKV), achieving 81% KV-cache reduction during inference. Despite its adoption in massive production...

https://arxiv.org/abs/2607.23054

38 CONF

Scoping Review of AI, Metrology, and ESG in the Semiconductor Sector: Implications for Safe and Sustainable by Design (SSbD)

ArXiv CS.AI 50%

arXiv:2607.23082v1 Announce Type: cross Abstract: The semiconductor sector faces a dual transition: scaling manufacturing execution through Artificial Intelligence (AI) while satisfying stringent sustainability mandates, such as the EU Carbon Border Adjustment Mechanism (CBAM)....

https://arxiv.org/abs/2607.23082

39 CONF

Poster: Rethinking Security in LLM Code Generation through Real-World Risk Scenarios

ArXiv CS.AI 50%

arXiv:2607.23088v1 Announce Type: cross Abstract: Large Language Models (LLMs) are widely used for code generation, yet their security behavior in realistic development workflows remains underexplored. Existing benchmarks often rely on explicitly specified security requirements,...

https://arxiv.org/abs/2607.23088

40 CONF

KAYROS: An Anytime and Exact Open-Source Solver for Duration-Minimization Time-Dependent Vehicle Routing. A Technical Report and a Case Study in Human-AI Engineering

ArXiv CS.AI 50%

arXiv:2607.23116v1 Announce Type: cross Abstract: KAYROS is an open-source solver for duration-minimization time-dependent vehicle routing problems, with or without time windows (TDVRPTW, TDVRP). In these variants, travel times change with departure time, and each route's...

https://arxiv.org/abs/2607.23116

41 CONF

Robustness and Cybersecurity in the EU Artificial Intelligence Act

ArXiv CS.AI 50%

arXiv:2502.16184v3 Announce Type: replace Abstract: The EU Artificial Intelligence Act (AIA) establishes different legal principles for different types of AI systems. While prior work has sought to clarify some of these principles, little attention has been paid to robustness...

https://arxiv.org/abs/2502.16184

42 CONF

Fashion-3DLR: A Controllable 3D Garment Generation Using Pairwise Fashion Elements for Intelligent Design

ArXiv CS.AI 50%

arXiv:2607.23189v1 Announce Type: cross Abstract: AI-generated content (AIGC) has made significant progress, with 2D generative models becoming ready-to-use tools for the digital fashion industry. However, 3D garment generation remains in its nascent stage, where in the realm of...

https://arxiv.org/abs/2607.23189

43 **CONF**

BoneAgeTW2: Automated Skeletal Maturation Assessment via the Tanner-Whitehouse 2 Method, Deep Learning, and Clinical Report Generation with Distribution Curves

ArXiv CS.AI 50%

arXiv:2607.23224v1 Announce Type: cross Abstract: We present BoneAgeTW2, the first fully open-source system to automate the complete Tanner-Whitehouse 2 (TW2) clinical protocol for skeletal maturity assessment end-to-end. The system employs YOLOv8 for precise detection and...

https://arxiv.org/abs/2607.23224

44 **CONF**

X-Stage: An Overlooked Pipeline Stage for Communication-Computation Overlap in DiT Inference

ArXiv CS.AI 50%

arXiv:2607.23264v1 Announce Type: cross Abstract: Fine-grained, device-initiated communication lets persistent GPU kernels in distributed diffusion transformer (DiT) inference issue remote stores and overlap data movement with Tensor Core computation. Existing systems schedule...

https://arxiv.org/abs/2607.23264

45 **CONF**

Statistically Supported LLM Ingredient and Recipe Data Collection in Computational Nutrition

ArXiv CS.AI 50%

arXiv:2607.23273v1 Announce Type: cross Abstract: Computational nutrition needs precise ingredient data, but current databases are incomplete, inconsistent, and built for human reference rather than automated reasoning. LLMs could help fill these gaps, but single-pass outputs...

https://arxiv.org/abs/2607.23273

46 **CONF**

Training with (Swap) Regret Loss in a Single-Layer Self-Attention Model: A Case Study on the Probability Simplex

ArXiv CS.AI 50%

arXiv:2607.23333v1 Announce Type: cross Abstract: We revisit the regret loss framework introduced in Park et al. (2025), which uses decision-theoretic regret as a direct loss function for training models to make better decisions, through the lens of probability-simplex policies....

https://arxiv.org/abs/2607.23333

47 **CONF**

Patient-Agnostic Synthetic Pretraining for Efficient Patient-Specific Intraoperative 2D/3D Registration

ArXiv CS.AI 50%

arXiv:2607.23343v1 Announce Type: cross Abstract: Intraoperative 2D/3D registration aligns preoperative CT volumes with intraoperative X-ray or fluoroscopic images and is essential for image-guided interventions. Recent learning-based and differentiable registration methods have...

https://arxiv.org/abs/2607.23343

48 **CONF**

On AI Safety and Security Technical Debt in Engineering AI-Enabled Systems

ArXiv CS.AI 50%

arXiv:2607.23365v1 Announce Type: cross Abstract: Artificial intelligence (AI) systems are increasingly deployed in high-stakes domains such as healthcare, autonomous driving, finance, and education. While these systems offer powerful data-driven and adaptive capabilities, their...

https://arxiv.org/abs/2607.23365

49 CONF

Fair Division with Strictly Increasing Valuations: A Tight Threshold for Two-Agent EF1 and PO

ArXiv CS.AI 50%

arXiv:2607.23367v1 Announce Type: cross Abstract: We study whether strictly positive marginal values restore the compatibility of envy-freeness up to one good (EF1) and Pareto optimality (PO) for indivisible goods. For two agents, we identify the exact threshold in the number of...

https://arxiv.org/abs/2607.23367

50 CONF

Explaining BiomedCLIP with Weighted Banzhaf Interactions Supported by Tree-Gram Parsing

ArXiv CS.AI 50%

arXiv:2607.23368v1 Announce Type: cross Abstract: Vision-Language Models (VLMs) are demonstrating significant capabilities in medical tasks like radiology analysis, yet providing faithful and interpretable explanations remains a key consideration for their responsible deployment...

https://arxiv.org/abs/2607.23368

51 CONF

When Activation Oracles Learn Not to Read: Concept-Specific Blind Spots in Fine-Tuned Oracles

ArXiv CS.AI 50%

arXiv:2607.23379v1 Announce Type: cross Abstract: Activation Oracles (AOs) are language models trained to answer natural-language questions about another model's internal activations. They offer a flexible interface for reading hidden information from model states, especially...

https://arxiv.org/abs/2607.23379

52 CONF

Directional Influence Function: Estimating Training Data Influence in Constrained Learning

ArXiv CS.AI 50%

arXiv:2607.23388v1 Announce Type: cross Abstract: As constrained learning becomes increasingly common, models are trained under explicit feasibility requirements to enforce fairness, safety, robustness, regulariza- tion, and physics or logic constraints. Understanding how...

https://arxiv.org/abs/2607.23388

53 CONF

Blood Pressure Estimation from PPG: A Comparative Study of Direct and ECG-Mediated Deep Learning Pipelines

ArXiv CS.AI 50%

arXiv:2607.23406v1 Announce Type: cross Abstract: Continuous cuffless blood pressure (BP) monitoring is essential for connected health systems and wearable devices, enabling early detection, longitudinal tracking, and personalized management of cardiovascular disease. Many prior...

https://arxiv.org/abs/2607.23406

54 CONF

Reasoning or Memorization: Can LLMs Understand and Generate Chinese Xiehouyu Riddles?

ArXiv CS.AI 50%

arXiv:2607.23440v1 Announce Type: cross Abstract: In this paper, we push the boundary of LLM reasoning by testing them in a Chinese language game, xiehouyu, with novel xiehouyu created by linguists that had not existed before to avoid data contamination. We use multiple-choice...

https://arxiv.org/abs/2607.23440

55 CONF

Do Small Models Use the Law You Give Them? Context-Injected Fine-Tuning for Legal QA in Bangladesh

ArXiv CS.AI 50%

arXiv:2607.23446v1 Announce Type: cross Abstract: A small language model can receive the governing statutory provision and still answer incorrectly. We test whether fine-tuning on examples containing relevant law improves later use of retrieved law. We curate 2{,}165 bilingual...

https://arxiv.org/abs/2607.23446

56 CONF

ATLAS: Automated Approximation of Transformers for Efficient Homomorphic Inference in One Hour

ArXiv CS.AI 50%

arXiv:2607.23478v1 Announce Type: cross Abstract: Fully homomorphic encryption (FHE) provides strong cryptographic guarantees for private inference, but deploying transformer models under FHE remains prohibitively expensive. A key bottleneck is that non-linear operations such as...

https://arxiv.org/abs/2607.23478

57 CONF

Novel Claim or D\'ej\`a Vu? Rethinking "Contamination-Free'' Dynamic Evaluation for Multimodal Automated Fact-Checking

ArXiv CS.AI 50%

arXiv:2607.23514v1 Announce Type: cross Abstract: Multimodal automated fact-checking (MAFC) verifies claims by retrieving and reasoning over external evidence. However, most existing static benchmarks risk contamination: they primarily consist of outdated claims verifiable using...

https://arxiv.org/abs/2607.23514

58 CONF

Mission-Level Runtime Assurance for LLM-Assisted ISR Swarms over a Verification-Aware Fabric

ArXiv CS.AI 50%

arXiv:2607.23532v1 Announce Type: cross Abstract: Swarms of LLM-assisted autonomous robots are increasingly proposed for cooperative intelligence, surveillance, and reconnaissance (ISR) in contested environments. A growing class of their assurance failures arises not within any...

https://arxiv.org/abs/2607.23532

59 CONF

DualityCert: Verifier-Gated Language-Model Repair of Broken Duality Claims in Quantum Field Theory

ArXiv CS.AI 50%

arXiv:2607.23614v1 Announce Type: cross Abstract: We present DualityCert, a symbolic verifier for candidate Seiberg-duality claims in four-dimensional N=1 quiver gauge theories. The verifier evaluates 't Hooft anomaly matching, superpotential R-charge consistency, central-charge...

https://arxiv.org/abs/2607.23614

60 CONF

Variational-Ising-Attention (VIA):TailoredAttentionMattersfor Science

ArXiv CS.AI 50%

arXiv:2607.23634v1 Announce Type: cross Abstract: Attention enables context modeling via query-key scoring with softmax normalization. Driven by industrial long-context demands, mainstream research has converged toward sparsity and efficiency--yet softmax's independence...

https://arxiv.org/abs/2607.23634

61 CONF

An empirical investigation into the properties of standard word embeddings

ArXiv CS.AI 50%

arXiv:2607.23675v1 Announce Type: cross Abstract: The embedding of word sequences into continuous vector spaces has been one of the most important developments in Natural Language Processing in the recent past. Such embeddings have found application in areas such as Automatic...

https://arxiv.org/abs/2607.23675

62 CONF

The Illusion of Secure LLM Code: Closing the Security Gap via Iterative Reprompting

ArXiv CS.AI 50%

arXiv:2607.23710v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly integrated into software development workflows, yet their ability to autonomously generate secure authentication code remains uncertain. This paper evaluates the security architecture...

https://arxiv.org/abs/2607.23710

63 CONF

AI Strategy: How to Choose What AI Product to Implement

ArXiv CS.AI 50%

arXiv:2607.23733v1 Announce Type: cross Abstract: Firms struggle to choose AI projects that pay off: two projects can look equally promising to smart, motivated stakeholders and yet deserve opposite decisions. At the residential real-estate brokerage Compass, one AI product...

https://arxiv.org/abs/2607.23733

64 CONF

WISERouter: LLM Routing with Workload Budget Constraint

ArXiv CS.AI 50%

arXiv:2607.23765v1 Announce Type: cross Abstract: Large language models (LLMs) achieve impressive performance across multiple domains, but using the most capable model for every query is prohibitive at scale. LLM routing exploits diversity in model capability and cost by...

https://arxiv.org/abs/2607.23765

65 CONF

Outcome-Fair Restless Multi-Armed Bandits for Stochastic Deadline Scheduling

ArXiv CS.AI 50%

arXiv:2607.23772v1 Announce Type: cross Abstract: We study a restless multi-armed bandit (RMAB) problem for a stochastic deadline scheduling application. RMAB problems are solved using the Whittle index policy. The goal in RMAB is to maximize the expected cumulative discounted...

https://arxiv.org/abs/2607.23772

66 CONF

Scale Weight Decay and Train Better

ArXiv CS.AI 50%

arXiv:2607.23777v1 Announce Type: cross Abstract: The discovery of scaling laws has motivated training neural networks on ever increasing quantities of data. This is typically done with a constant decoupled weight decay which causes the network weights to shrink steadily over...

https://arxiv.org/abs/2607.23777

67 CONF

StanceFlip: A Comprehensive Multi-Dimensional Benchmark for Multimodal Conversational Stance Flipping Forecasting

ArXiv CS.AI 50%

arXiv:2607.24191v1 Announce Type: cross Abstract: Conversational stance detection has shifted from static text analysis to dynamic multimodal modeling. However, existing benchmarks exhibit three key limitations: failure to capture the dynamic evolution of beliefs, particularly...

https://arxiv.org/abs/2607.24191

68 CONF

Kalypso: Relational LLM Serving

ArXiv CS.AI 50%

arXiv:2607.23815v1 Announce Type: cross Abstract: Large language models are increasingly used as semantic operators for filtering, extracting, ranking, joining, and transforming unstructured data. Existing semantic query processing systems invoke request-centric LLM serving...

https://arxiv.org/abs/2607.23815

69 CONF

MulRobBench: A Decision-Level Benchmark for Safe and Security-Policy-Compliant Multimodal UAV Agents

ArXiv CS.AI 50%

arXiv:2607.23870v1 Announce Type: cross Abstract: Smart-city airspace is transforming Uncrewed Aerial Vehicles (UAVs) from passive sensing platforms into cyber-physical decision makers that must follow operational rules under degraded observations and ambiguous language....

https://arxiv.org/abs/2607.23870

70 CONF

Harnessing X-ray Absorption Spectroscopy Data through Multimodal Mining of Battery Literature

ArXiv CS.AI 50%

arXiv:2607.23886v1 Announce Type: cross Abstract: X-ray absorption spectroscopy (XAS) is central to understanding the local electronic and atomic structure of materials, yet most published spectra remain inaccessible to data-driven analysis because they are embedded in figures...

https://arxiv.org/abs/2607.23886

71 CONF

Visible to the Court: How AI Is (and Isn't) Litigated in U.S. Federal Court Opinions

ArXiv CS.AI 50%

arXiv:2607.23888v1 Announce Type: cross Abstract: In the United States, artificial intelligence (AI) is rapidly deployed amid limited federal regulation. With courts become a recurring forum in which AI-related practices are scrutinized, it is important to empirically understand...

https://arxiv.org/abs/2607.23888

72 CONF

Embodied GPT-5.1: Evidence of a World Model?

ArXiv CS.AI 50%

arXiv:2607.23899v1 Announce Type: cross Abstract: This exploratory study examines whether a large multimodal language model, GPT-5.1, can serve as the high-level controller of a physical mobile robot despite having no prior embodiment, no training in simulated environments, and...

https://arxiv.org/abs/2607.23899

73 **CONF**

DuoAD: Leveraging [CLS] Dual Characteristics for Training-Free Few-Shot Anomaly Detection

ArXiv CS.AI **50%**

arXiv:2607.23924v1 Announce Type: cross Abstract: Vision foundation models have enabled strong training-free anomaly detection (AD). However, most existing approaches rely primarily on independent local patch features, leaving the global contextual information encoded by Vision...

https://arxiv.org/abs/2607.23924

74 **CONF**

HELIOS: An LLM-Driven Autonomous Indirect Trajectory Optimization Agent

ArXiv CS.AI **50%**

arXiv:2607.24051v1 Announce Type: cross Abstract: Low-thrust trajectory optimization is a core technology in deep-space mission design. Indirect methods based on Pontryagin's Minimum Principle (PMP) offer rigorous optimality guarantees, yet their practical application faces...

https://arxiv.org/abs/2607.24051

75 **CONF**

Capacity-Aware Deep Learning for Generalizable Traffic Volume Estimation Across Links and Cities

ArXiv CS.AI **50%**

arXiv:2607.24056v1 Announce Type: cross Abstract: Network-wide traffic volume estimation typically relies on propagating measurements from fixed sensors, making performance highly dependent on sensor density and limiting deployment in sparsely instrumented networks. We propose a...

https://arxiv.org/abs/2607.24056

76 **CONF**

Towards simultaneous decoding of kinetic and kinematic movement parameters during grasp and lift task by noninvasive brain imaging

ArXiv CS.AI **50%**

arXiv:2607.24081v1 Announce Type: cross Abstract: Brain-machine interfaces (BMIs) can assist individuals with limited mobility, such as stroke survivors or amputees. One of the key challenges in developing BMIs is expanding their usability and control, which can be achieved by...

https://arxiv.org/abs/2607.24081

77 **CONF**

LU-500: A Logo Benchmark for Concept Unlearning

ArXiv CS.AI **50%**

arXiv:2607.24101v1 Announce Type: cross Abstract: Concept unlearning is increasingly used to limit the reproduction of protected or unsafe visual concepts in text-to-image models. Existing evaluations, however, mostly study targets that dominate the whole image, such as styles,...

https://arxiv.org/abs/2607.24101

78 **CONF**

Physics-Guided Generative AI for Property-Targeted 3D Porous Media Design

ArXiv CS.AI **50%**

arXiv:2607.24274v1 Announce Type: cross Abstract: Inverse design of three-dimensional porous media is central to applications in filtration, catalysis, energy storage, fuel cells, thermal management, and biomedical scaffolds, but remains challenging because many distinct pore...

https://arxiv.org/abs/2607.24274

79 CONF

The Tokenizer Tax: Quantifying and Explaining the Cross-Lingual Cost of Subword Tokenization for Indian Languages

ArXiv CS.AI 50%

arXiv:2607.24276v1 Announce Type: cross Abstract: Large language models (LLMs) process text through subword tokenizers rather than directly reading characters or words. Because these tokenizers are trained predominantly on English-centric corpora, they introduce a systematic and...

https://arxiv.org/abs/2607.24276

80 CONF

Teacher Knows It Best: Spontaneous Symmetry Breaking and Tipping Points in Networked Langevin Dynamics AI Sycophancy

ArXiv CS.AI 50%

arXiv:2607.24304v1 Announce Type: cross Abstract: We formulate a statistical physics framework to model a networked stochastic dynamical system exhibiting bistability, driven by additive noise and social conformity. We apply this model to understand and mitigate AI-induced...

https://arxiv.org/abs/2607.24304

81 CONF

Beyond Aggregate Risk: Role-Stratified Conformal Risk Control for LLM Tool Calls

ArXiv CS.AI 50%

arXiv:2607.24343v1 Announce Type: cross Abstract: Language-model agents act through structured tool calls whose arguments carry different risks. Untrusted content may safely influence an email body but should not determine a recipient, account, command, or credential. Existing...

https://arxiv.org/abs/2607.24343

82 CONF

Closed-Loop Validation-Repair for Healthcare Interoperability: A Multi-Model Study of Schema Compliance in Clinical LLMs

ArXiv CS.AI 50%

arXiv:2607.24371v1 Announce Type: cross Abstract: Healthcare interoperability requires AI systems to produce structured outputs conforming to standardized schemas including ICD-10 for diagnostic coding, CPT for procedure billing, and HL7 FHIR for data exchange. While large...

https://arxiv.org/abs/2607.24371

83 CONF

Regulating for AI Legitimacy

ArXiv CS.AI 50%

arXiv:2607.24391v1 Announce Type: cross Abstract: AI systems already govern. They rank speech and allocate attention, filter applicants and triage claims. The dominant frame for AI governance, alignment, asks whether such systems pursue the right objectives safely. It cannot...

https://arxiv.org/abs/2607.24391

84 CONF

The SpiNNaker2 chip: a many-core platform for flexible and scalable brain-inspired computing

ArXiv CS.AI 50%

arXiv:2607.24396v1 Announce Type: cross Abstract: In deep learning, efficiency gets more and more important to compensate for the ongoing growth in model sizes and applications. Neuromorphic hardware has long been advocated as an upcoming alternative to deep networks, taking...

https://arxiv.org/abs/2607.24396

85 CONF

DraftExpert: Expansion-Aware Self-Speculative Decoding for End-Device MoE Inference

ArXiv CS.AI 50%

arXiv:2607.24434v1 Announce Type: cross Abstract: Large Mixture-of-Experts (MoE) language models are attractive for end-device deployment because only a small subset of experts is active per token, but their routed expert weights often exceed accelerator memory. We target...

https://arxiv.org/abs/2607.24434

86 CONF

Seesaw: Accelerating Training by Balancing Learning Rate and Batch Size Scheduling

ArXiv CS.AI 50%

arXiv:2510.14717v2 Announce Type: replace-cross Abstract: Increasing the batch size during training -- a ''batch ramp'' -- is a promising strategy to accelerate large language model pretraining. While for SGD, doubling the batch size can be equivalent to halving the learning...

https://arxiv.org/abs/2510.14717

87 CONF

UNIFUSION: Adapting Autoregressive Language Models into Discrete Diffusion under a Unified Reverse-Rate Objective

ArXiv CS.AI 50%

arXiv:2607.24507v1 Announce Type: cross Abstract: Existing methods mainly adapt pretrained autoregressive (AR) language models to masked diffusion, whereas we directly adapt them to uniform-noise diffusion, where every token remains editable during sampling. However, adapting AR...

https://arxiv.org/abs/2607.24507

88 CONF

EchoBridge: Long-Tail-Aware ECG-Echocardiography Text Alignment for Echocardiography-Derived Cardiac Findings

ArXiv CS.AI 50%

arXiv:2607.24553v1 Announce Type: cross Abstract: Standardized echocardiography conclusions provide meaningful supervision for learning ECG representations of echocardiography-derived cardiac findings. Global ECG--text alignment may entangle modality-specific factors, while...

https://arxiv.org/abs/2607.24553

89 CONF

The Visual Bottleneck: Sparse-Frame Adaptation of MLLMs for Joint Spatial-Temporal Video Grounding

ArXiv CS.AI 50%

arXiv:2607.24570v1 Announce Type: cross Abstract: Large-scale video platforms process millions of uploads hourly, requiring moderation systems that can localize when and where policy violations occur within each video. Processing every frame is infeasible at scale, so systems...

https://arxiv.org/abs/2607.24570

90 CONF

Looping Is Not Reliability: State-Bound Evidence and Typed Revision Contracts for Agentic Code Repair

ArXiv CS.AI 50%

arXiv:2607.24604v1 Announce Type: cross Abstract: Generate--test--revise loops are common in coding agents, but repetition alone provides no reliability guarantee. We study the gap between finding a correct patch and retaining, verifying, and submitting it. A sealed five-seed...

https://arxiv.org/abs/2607.24604

91 CONF

A corrective agentic hybrid RAG and an operations-grounded evaluation for a scientific facility

ArXiv CS.AI 50%

arXiv:2607.24663v1 Announce Type: cross Abstract: Scientific user facilities accumulate decades of operational knowledge that no single search index covers: electronic logbooks, technical documents, internal wikis, operations chat messages, maintenance records, and live...

https://arxiv.org/abs/2607.24663

92 CONF

Denial of Deadline: Network-Driven Accuracy Collapse in Distributed Inference Pipelines

ArXiv CS.AI 50%

arXiv:2607.24692v1 Announce Type: cross Abstract: Inference systems increasingly combine a fast path that returns predictions within the application's latency deadline together with a higher-accuracy slow path that runs higher-compute methods on stronger, remote hardware, so its...

https://arxiv.org/abs/2607.24692

93 CONF

Efficient LLM-Generated Shuttling Compilers for Complex Trapped-Ion Architectures

ArXiv CS.AI 50%

arXiv:2607.24714v1 Announce Type: cross Abstract: Trapped-ion quantum computers rely on shuttling compilers, which cast an input algorithm into a sequence of ion-qubit movements within a given architecture. We present the first study in which a single frontier large language...

https://arxiv.org/abs/2607.24714

94 CONF

The Physics of Multi-Turn Long-Horizon Planning: From Pre-training to Post-training via Single- and Multi-Teacher On-Policy Agentic Distillation

ArXiv CS.AI 50%

arXiv:2607.24720v1 Announce Type: cross Abstract: Multi-turn long-horizon planning is critical for foundation model agents, yet how to fundamentally improve it remains unclear. Existing models are trained on uncontrollable and opaque Internet data, making it difficult to...

https://arxiv.org/abs/2607.24720

95 CONF

KANEx: Translating Kolmogorov-Arnold Networks' Interpretability to Medical Explainability

ArXiv CS.AI 50%

arXiv:2607.24730v1 Announce Type: cross Abstract: Computer vision models have become highly effective for medical applications, yet their black-box nature continues to undermine clinician trust. In clinical workflows, chest X-ray classifiers are increasingly paired with...

https://arxiv.org/abs/2607.24730

96 CONF

Rethinking Classifier-Free Guidance in On-Policy Diffusion Distillation

ArXiv CS.AI 50%

arXiv:2607.24731v1 Announce Type: cross Abstract: On-policy distillation (OPD) adapts diffusion models by querying a teacher along trajectories generated by the current student, but how it should behave under classifier-free guidance (CFG), a default component of modern...

https://arxiv.org/abs/2607.24731

97 **CONF**

ClinFusion: A Vision-Centric Multimodal LLM System for Holistic Medical Understanding

ArXiv CS.AI 50%

arXiv:2607.24743v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) hold immense potential to revolutionize clinical practice, yet deploying them in the medical domain is fundamentally a vision-centric challenge: models must absorb knowledge from...

https://arxiv.org/abs/2607.24743

98 **CONF**

ANSR-DT: A Neuro-Symbolic Framework for Adaptive and Explainable Digital Twins

ArXiv CS.AI 50%

arXiv:2501.08561v5 Announce Type: replace Abstract: Digital twins are increasingly used to monitor and optimize industrial systems, yet many existing frameworks remain difficult to interpret, slow to adapt, and limited in their ability to incorporate explicit domain knowledge....

https://arxiv.org/abs/2501.08561

99 **CONF**

Hybrid AI-Physical Modeling for Penetration Bias Correction in X-band InSAR DEMs: A Greenland Case Study

ArXiv CS.AI 50%

arXiv:2504.08909v2 Announce Type: replace Abstract: Digital elevation models derived from Interferometric Synthetic Aperture Radar (InSAR) data over glacial and snow-covered regions often exhibit systematic elevation errors, commonly termed "penetration bias." We leverage...

https://arxiv.org/abs/2504.08909

00 **CONF**

What Does Chain-of-Thought Contribute at Probe Time? Evidence for Local Co-Occurrence Activation

ArXiv CS.AI 50%

arXiv:2605.26795v2 Announce Type: replace Abstract: Chain-of-thought (CoT) prompting enhances large language model performance, yet what drives these gains remains unclear. We study this question from a probe-time perspective: holding CoT rationales fixed, we test which textual...

https://arxiv.org/abs/2605.26795

01 **CONF**

PhantomFill: When the Form Demands an Answer, Language Models Invent One

ArXiv CS.AI 50%

arXiv:2607.20492v2 Announce Type: replace-cross Abstract: Language models in production do not write prose. They fill forms: JSON fields, function arguments, extraction templates. We show that the form itself causes hallucination. We ask thirteen models the same question about...

https://arxiv.org/abs/2607.20492

02 **CONF**

The Hitchhiker's Guide to Agentic AI: From Foundations to Systems

ArXiv CS.AI 50%

arXiv:2606.24937v2 Announce Type: replace Abstract: The Hitchhiker's Guide to Agentic AI is a comprehensive practitioner's reference for building autonomous AI systems. The book covers the full stack from first principles to production deployment, organized around a central...

https://arxiv.org/abs/2606.24937

03 CONF

Athena-Brain Technical Report: An Efficient Robot Brain for General Intelligence and Embodied Interaction

ArXiv CS.AI 50%

arXiv:2607.18985v2 Announce Type: replace Abstract: Large language models (LLMs) have demonstrated remarkable capabilities in language understanding, reasoning, and world knowledge. As embodied agents become increasingly capable, there is a growing demand for compact models that...

https://arxiv.org/abs/2607.18985

04 CONF

TRACE-ROUTER: Task-Consistent and Adaptive Online Routing for Agentic AI

ArXiv CS.AI 50%

arXiv:2607.22465v2 Announce Type: replace Abstract: Routing to select large language models (LLMs) with different cost-quality trade-offs has become a fundamental deployment feature of enterprise AI. Existing routers, primarily make independent routing decisions for each LLM...

https://arxiv.org/abs/2607.22465

05 CONF

Beyond Squared Error: Exploring Loss Design for Enhanced Training of Generative Flow Networks

ArXiv CS.AI 50%

arXiv:2410.02596v2 Announce Type: replace-cross Abstract: Generative Flow Networks (GFlowNets) are a novel class of generative models designed to sample from unnormalized distributions and have found applications in various important tasks, attracting great research interest in...

https://arxiv.org/abs/2410.02596

06 CONF

Sign-Symmetry Learning Rules are Robust Fine-Tuners

ArXiv CS.AI 50%

arXiv:2502.05925v2 Announce Type: replace-cross Abstract: Backpropagation (BP) has long been the predominant method for training neural networks due to its effectiveness. However, numerous alternative approaches, broadly categorized under feedback alignment, have been proposed,...

https://arxiv.org/abs/2502.05925

07 CONF

A Survey of Graph Transformers: Architectures, Theories and Applications

ArXiv CS.AI 50%

arXiv:2502.16533v3 Announce Type: replace-cross Abstract: Graph Transformers (GTs) have demonstrated a strong capability in modeling graph structures by addressing the intrinsic limitations of graph neural networks (GNNs), such as over-smoothing and over-squashing. Recent...

https://arxiv.org/abs/2502.16533

08 CONF

Kinship Verification through a Forest Neural Network

ArXiv CS.AI 50%

arXiv:2504.18910v2 Announce Type: replace-cross Abstract: Early methods used face representations in kinship verification, which are less accurate than joint representations of parents' and children's facial images learned from scratch. We propose an approach featuring graph...

https://arxiv.org/abs/2504.18910

09 CONF

Steerable Chatbots: Exploring Personalization Control Interfaces via LLM Activation Steering

ArXiv CS.AI 50%

arXiv:2505.04260v3 Announce Type: replace-cross Abstract: Personalizing LLM responses typically requires users to articulate their preferences through prompting, which can be burdensome at cold start and difficult to articulate in natural language. We introduce an alternative...

https://arxiv.org/abs/2505.04260

10 CONF

LEDOM: Reverse Language Model

ArXiv CS.AI 50%

arXiv:2507.01335v4 Announce Type: replace-cross Abstract: Autoregressive language models are trained exclusively left-to-right. We explore the complementary factorization, training right-to-left at scale, and ask what reasoning patterns emerge when a model conditions on future...

https://arxiv.org/abs/2507.01335

11 CONF

PrinciplismQA: A Philosophy-Grounded Approach to Assessing LLM-Human Clinical Medical Ethics Alignment

ArXiv CS.AI 50%

arXiv:2508.05132v3 Announce Type: replace-cross Abstract: As medical LLMs transition to clinical deployment, assessing their ethical reasoning capability becomes critical. While achieving high accuracy on knowledge benchmarks, LLMs lack validated assessment for navigating...

https://arxiv.org/abs/2508.05132

12 CONF

OpenAIs HealthBench in Action: Evaluating an LLM-Based Medical Assistant on Realistic Clinical Queries

ArXiv CS.AI 50%

arXiv:2509.02594v3 Announce Type: replace-cross Abstract: Evaluating large language models (LLMs) on their ability to generate high-quality, accurate, situationally aware answers to clinical questions requires going beyond conventional benchmarks to assess how these systems...

https://arxiv.org/abs/2509.02594

13 CONF

Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers

ArXiv CS.AI 50%

arXiv:2509.03059v2 Announce Type: replace-cross Abstract: Recent advances in Large Language Models (LLMs) have shown that their reasoning capabilities can be significantly improved through Reinforcement Learning with Verifiable Reward (RLVR), particularly in domains like...

https://arxiv.org/abs/2509.03059

14 CONF

A funny companion: Distinct neural responses to AI- versus human-attributed humor

ArXiv CS.AI 50%

arXiv:2509.10847v3 Announce Type: replace-cross Abstract: As artificial intelligence (AI) companions become capable of human-like communication, including telling jokes, understanding how people cognitively and affectively respond to AI-attributed humor becomes increasingly...

https://arxiv.org/abs/2509.10847

15 CONF

VLASH: Real-Time VLAs via Future-State-Aware Asynchronous Inference

ArXiv CS.AI 50%

arXiv:2512.01031v2 Announce Type: replace-cross Abstract: Vision-Language-Action models (VLAs) are becoming increasingly capable across diverse robotic tasks. However, these models are typically deployed under synchronous inference, where the robot waits for model inference to...

https://arxiv.org/abs/2512.01031

16 CONF

GFLAN: Generative Functional Layouts

ArXiv CS.AI 50%

arXiv:2512.16275v2 Announce Type: replace-cross Abstract: Automated floor plan generation lies at the intersection of combinatorial search, geometric constraint satisfaction, and functional design requirements -- a confluence that has historically resisted a unified...

https://arxiv.org/abs/2512.16275

17 CONF

TextBridgeGNN: Pre-training Graph Neural Network for Cross-Domain Recommendation via Text-Guided Transfer

ArXiv CS.AI 50%

arXiv:2601.02366v3 Announce Type: replace-cross Abstract: Graph-based recommendation has achieved great success in recent years. The classical graph recommendation model utilizes ID embedding to store essential collaborative information. However, this ID-based paradigm faces...

https://arxiv.org/abs/2601.02366

18 CONF

Infinite-Precision Autoregressive Modeling for Vector Graphics and Layouts

ArXiv CS.AI 50%

arXiv:2601.05680v2 Announce Type: replace-cross Abstract: While Transformer-based autoregressive models excel in data generation, their token discretization strategy inherently limits their precision in continuous domains. We analyze the scalability limitations of existing...

https://arxiv.org/abs/2601.05680

19 CONF

UP-Fuse: Uncertainty-guided LiDAR-Camera Fusion for 3D Panoptic Segmentation

ArXiv CS.AI 50%

arXiv:2602.19349v2 Announce Type: replace-cross Abstract: LiDAR-camera fusion enhances 3D panoptic segmentation by leveraging camera images to complement sparse LiDAR scans, but it also introduces a critical failure mode. Under adverse conditions, degradation or failure of the...

https://arxiv.org/abs/2602.19349

20 CONF

How College Students Use AI to Navigate Course Readings: Evidence from an Eight-Week Study

ArXiv CS.AI 50%

arXiv:2602.09907v3 Announce Type: replace-cross Abstract: College students increasingly use AI chatbots to support academic reading, yet we lack granular understanding of how these interactions shape their reading experience and cognitive engagement. We conducted an eight-week...

https://arxiv.org/abs/2602.09907

21 CONF

LLM-Based Scientific Equation Discovery via Physics-Informed Token-Regularized Policy Optimization

ArXiv CS.AI 50%

arXiv:2602.10576v2 Announce Type: replace-cross Abstract: Symbolic regression aims to distill mathematical equations from observational data. Recent approaches have successfully leveraged Large Language Models (LLMs) to generate equation hypotheses, capitalizing on their vast...

https://arxiv.org/abs/2602.10576

22 CONF

The Equalizer: Introducing Shape-Gain Decomposition in Neural Audio Codecs

ArXiv CS.AI 50%

arXiv:2602.15491v2 Announce Type: replace-cross Abstract: Neural audio codecs (NACs) typically encode the short-term energy (gain) and normalized structure (shape) of speech/audio signals jointly within the same latent space. As a result, they are poorly robust to a global...

https://arxiv.org/abs/2602.15491

23 CONF

AIFL: A Global Daily Streamflow Forecasting Model Using a Deterministic LSTM Pre-trained on ERA5-Land and Fine-tuned on IFS

ArXiv CS.AI 50%

arXiv:2602.16579v2 Announce Type: replace-cross Abstract: Reliable global streamflow forecasting is essential for flood preparedness and water resource management, yet data-driven models often suffer from a performance gap when transitioning from historical reanalysis to...

https://arxiv.org/abs/2602.16579

24 CONF

From "Help" to Helpful: A Hierarchical Assessment of LLMs in Mental e-Health Applications

ArXiv CS.AI 50%

arXiv:2602.18443v2 Announce Type: replace-cross Abstract: Psychosocial online counselling frequently encounters generic subject lines that impede efficient case prioritisation. This study evaluates eleven large language models generating six-word subject lines for German...

https://arxiv.org/abs/2602.18443

25 CONF

Exact and Asymptotically Complete Robust Verifications of Neural Networks via Ising Solvers

ArXiv CS.AI 50%

arXiv:2603.00408v2 Announce Type: replace-cross Abstract: We present an Ising-compatible framework for formal neural-network robustness verification under bounded input perturbations. For piecewise-linear activations, the Exact Logarithmic PWL Model (Log-PWL) provides an exact,...

https://arxiv.org/abs/2603.00408

26 CONF

Statistical realism is not evidence that LLMs can estimate treatment effects in social science experiments

ArXiv CS.AI 50%

arXiv:2604.02458v3 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used to simulate human responses and estimate treatment effect of interventions when real-world experiments are costly or infeasible. The treatment-effect estimates are often...

https://arxiv.org/abs/2604.02458

27 **CONF**

How Transformers Learn to Plan via Multi-Token Prediction

ArXiv CS.AI 50%

arXiv:2604.11912v2 Announce Type: replace-cross Abstract: While next-token prediction (NTP) has been the standard objective for training language models, it often struggles to capture global structure in reasoning tasks. Multi-token prediction (MTP) has recently emerged as a...

https://arxiv.org/abs/2604.11912

28 **CONF**

Adaptive receptive field-based spatial-frequency feature reconstruction network for fine-grained few-shot image classification

ArXiv CS.AI 50%

arXiv:2604.16936v2 Announce Type: replace-cross Abstract: Feature reconstruction techniques are widely applied for few-shot fine-grained image classification (FSFGIC). Our research indicates that one of the main challenges facing existing feature-based FSFGIC methods is how to...

https://arxiv.org/abs/2604.16936

29 **CONF**

Principles and Guidelines for Randomized Controlled Trials in AI Evaluation

ArXiv CS.AI 50%

arXiv:2605.02050v2 Announce Type: replace-cross Abstract: This work establishes a framework for standardizing AI evaluation RCTs (sometimes called human uplift studies). Drawing on established practices from disciplines with established RCT traditions, including software...

https://arxiv.org/abs/2605.02050

30 **CONF**

SwitchBraidNet: Quantisation-Aware Lightweight Architecture for Hybrid Brain-Computer Interface

ArXiv CS.AI 50%

arXiv:2606.18816v2 Announce Type: replace-cross Abstract: Hybrid brain-computer interfaces (BCIs) that integrate motor imagery (MI) and steady-state visual evoked potentials (SSVEP) provide high-dimensional neural decoding but typically exceed the computational limits of...

https://arxiv.org/abs/2606.18816

31 **CONF**

TextRich: A Multi-Domain Benchmark for Detecting AI-Generated Text-Rich Images from GPT-Image-2

ArXiv CS.AI 50%

arXiv:2606.19259v2 Announce Type: replace-cross Abstract: Text-rich images often contain privacy-sensitive, transactional, or decision-relevant information. As recent multimodal image generation models become increasingly capable of synthesizing realistic textual content and...

https://arxiv.org/abs/2606.19259

32 **CONF**

EmotionAI: A Privacy-Preserving Computational Intelligence Pipeline for Speech-Emotion-Grounded Conversational Analysis

ArXiv CS.AI 50%

arXiv:2606.24941v2 Announce Type: replace-cross Abstract: Reviewing recorded interviews for affective cues such as composure and agitation is slow and subjective, and cloud services that could automate the task require sensitive audio to leave the device. EmotionAI is a fully...

https://arxiv.org/abs/2606.24941

33 CONF

QuantFlow: A Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting

ArXiv CS.AI 50%

arXiv:2607.02632v2 Announce Type: replace-cross Abstract: Time-series forecasting supports decisions in finance, en-ergy, transportation, public health, and industrial monitoring. Recent foundation models improve transfer across forecast-ing tasks, but many depend on centralized...

https://arxiv.org/abs/2607.02632